EX 7 – Demonstration of Linear and Logistic regression

AIM :

To demonstrate Linear Regression and Logistic Regression using datasets from different domains, and analyze their performance for prediction tasks.

ALGORITHM :

1. Data Preparation

    1. Import required libraries (Pandas, NumPy, Scikit-learn)
    2. Load the dataset (e.g., housing, medical, or classification dataset)
    3. Handle missing values and preprocess data
    4. Split data into training and testing sets

2. Linear Regression

(Used for continuous output prediction)

    1. Select dependent and independent variables
    2. Train the Linear Regression model on training data
    3. Predict values using test data
    4. Evaluate model using metrics like:
        - Mean Squared Error (MSE)
        - R^2 Score
    5. Visualize regression line (optional)

3. Logistic Regression

(Used for classification problems)

    1. Select features and target variable (categorical)
    2. Train the Logistic Regression model
    3. Predict class labels for test data
    4. Evaluate model using:
        - Accuracy
        - Confusion Matrix
        - Precision and Recall

CODE :

```
import warnings
warnings.filterwarnings("ignore")

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import mean_squared_error, accuracy_score

df = pd.read_csv("StudentsPerformance.csv")

X = df[["reading score","writing score"]]
y = df["math score"]

X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2)

model = LinearRegression()
model.fit(X_train,y_train)

pred = model.predict(X_test)

print(mean_squared_error(y_test,pred))

df["pass"] = (df["math score"] >= 50).astype(int)

X = df[["reading score","writing score"]]
y = df["pass"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2)

log = LogisticRegression(max_iter=500)
log.fit(X_train,y_train)

pred = log.predict(X_test)

print(accuracy_score(y_test,pred))
```

OUTPUT :

74.49667524165562

0.895

RESULT :

Thus the above program is executed and the result is obtained.